Bode Plots and Feedback

1. Bode Plots
   - an averaging filter
   - polar and Bode plots
2. Aliasing and Decibels
   - aliasing
   - decibels
3. Feedback
   - controlling a process
   - stabilize an unstable plant

MCS 472 Lecture 8
Industrial Math & Computation
Jan Verschelde, 30 January 2026

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an averaging filter

$$\xrightarrow{\quad u_{k-1} \ u_k \quad} \boxed{F} \xrightarrow{\quad y_k = (u_k + u_{k-1})/2 \quad}$$

For input $u = \{u_0, u_1, u_2, \ldots\}$, the output is

$$y = \{u_0/2, (u_0 + u_1)/2, (u_1 + u_2)/2, \ldots\}.$$

The impulse response determines a linear, time invariant, causal filter.

$$\xrightarrow{\quad 1 \quad 0 \quad 0 \quad \cdots \quad} \boxed{F} \xrightarrow{\quad 1/2 \quad 1/2 \quad 0 \quad \cdots \quad}$$

The impulse response is finite. The averaging filter is an example of a *Finite Impulse Response* (FIR) filter.

the transfer function of an averaging filter

$$\xrightarrow{\;1 \quad 0 \quad 0 \quad \cdots\;} \boxed{F} \xrightarrow{\;1/2 \quad 1/2 \quad 0 \quad \cdots\;}$$

The input $u = \{u_0, u_1, u_2, \ldots\}$ has z-transform $U(z) = \sum_{k=0}^{\infty} u_k z^{-k}$.

The output y has z-transform

$$Y(z) = (U(z) + z^{-1} U(z))/2 = \left(\frac{1 + z^{-1}}{2} \right) U(z).$$

The filter F has transfer function $H(z) = (1 + z^{-1})/2$.

What is the effect of this filter on signals?

amplitude gain and phase shift

we proved this theorem in lecture 7

Theorem

Let $H(z) = \sum_{k=0}^{\infty} h_k z^{-k}$ *be the transfer function of a filter* F*.*

For input $u = \left\{ u_k = \sin(\omega k T) \right\}_{k=0}^{\infty}$, $y = \left\{ y_k = r \sin(\omega k T + \phi) \right\}_{k=0}^{\infty}$ *is the output, where*

- $r = |H(e^{i\omega T})|$ *is the amplitude gain, and*
- $\phi = \arg H(e^{i\omega T})$ *is the phase shift.*

Interpret the input u as

- $u = \sin(\omega t)$, $\omega = 2\pi n$, at n Hertz,
- sampled at $t = kT$, T is the sampling rate.

Complex numbers appear, the imaginary unit is $i = \sqrt{-1}$.

amplitude gain and phase shift for the averaging filter

For the averaging filter $H(z) = (1 + z^{-1})/2$, compute

- $r = |H(e^{i\omega T})|$ the amplitude gain, and
- $\phi = \arg H(e^{i\omega T})$ the phase shift.

For $u = \sin(\omega t)$, $\omega = 2\pi n$, sampled at $t = kT$:

$$\begin{aligned} H(e^{i\omega T}) &= (1 + e^{-i\omega T})/2 \\ &= (1 + \cos(\omega T))/2 - i \sin(\omega T)/2 \end{aligned}$$

For $x = a + bi$, $|x| = \sqrt{a^2 + b^2}$ and $\arg(x) = \arctan(b/a)$.

The amplitude gain and phase shift are then

$$r = \frac{\sqrt{1 + \cos(\omega T)}}{\sqrt{2}} \quad \text{and} \quad \phi = -\arctan\left(\frac{\sin(\omega T)}{1 + \cos(\omega T)}\right).$$

frequency and sampling rate

For $\omega = 2\pi n$, n is the frequency, and sampling rate T:

$$H(e^{i\omega T}) = (1 + \cos(\omega T))/2 - i \sin(\omega T)/2,$$

the amplitude gain and phase shift for the averaging filter are

$$r = \frac{\sqrt{1 + \cos(\omega T)}}{\sqrt{2}} \quad \text{and} \quad \phi = -\arctan\left(\frac{\sin(\omega T)}{1 + \cos(\omega T)}\right).$$

Do the formulas answer our original question:

What is the effect of this filter on signals?

To interpret the formulas, we make a plot:

1. Fix the sampling rate T at 1000 samples per second.
2. Let the frequencies n range from 0 to 1000.

stability

A stable filter produces bounded outputs from bounded inputs.

Exercise 1:

Consider a filter with transfer function

$$H(z) = \frac{az + b}{(z - \alpha_1)(z - \alpha_2)}, \quad \alpha_1 \neq \alpha_2.$$

Show that the filter is stable if and only if $|\alpha_1| < 1$ and $|\alpha_2| < 1$.

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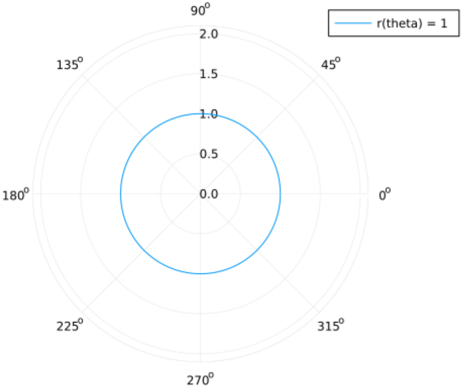
the polar plot of the unit circle

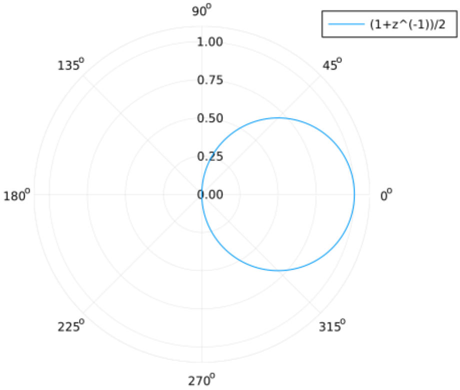
In polar coordinates, the unit circle is represented as

$$r(\theta) = 1, \quad \theta \in [0, 2\pi[.$$



polar plot of the averaging filter

$$r = \frac{\sqrt{1+\cos(\omega T)}}{\sqrt{2}} \quad \text{and} \quad \phi = -\arctan\left(\frac{\sin(\omega T)}{1+\cos(\omega T)}\right).$$



the Bode plot

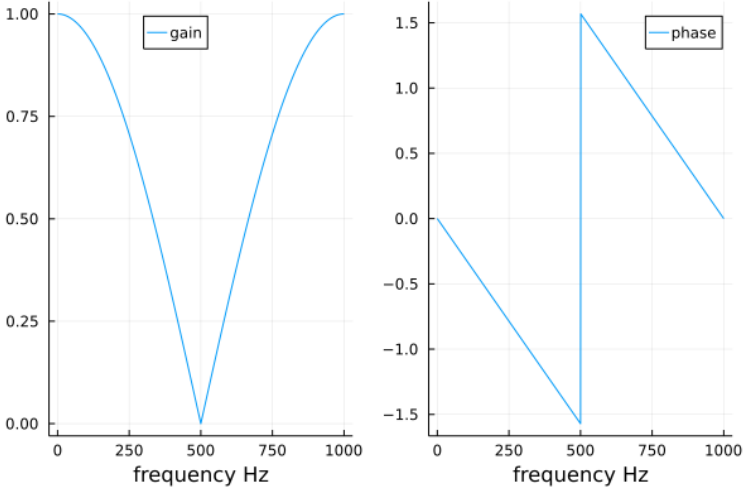
In a Bode plot, the amplitude gain and phase shift are separated.

The code with `Plots.jl` is below.

```
T = 1/1000
freq = 0:1:1000
omega = [2*pi*n for n in freq]
rw = [sqrt(1 + cos(w*T))/sqrt(2) for w in omega]
phi = [-atan(sin(w*T)/(1+cos(w*T))) for w in omega]
gainplot = plot(freq, rw, label="gain")
phaseplot = plot(freq, phi, label ="phase")
plot(gainplot, phaseplot)
```

See the posted Jupyter notebook.

the Bode plot of the averaging filter



What is the effect? *Higher frequencies are attenuated.*
This is the *lowpass* characteristic of this filter.

removing high frequency noise

Exercise 2:
Let $H(z)$ be the transfer function of a filter
to remove a signal at 60 Hz, sampled at 720 Hz.

(See Exercise 3 of L-7.)

1. Make the Bode plot for this filter.
2. Use the Bode plot to explain which signals are also removed by this filter.

Bode Plots and Feedback

aliasing

Consider:

$$H(e^{i\omega T}) = re^{i\phi} = H(e^{i(\omega T + 2\pi)}).$$

The sampled filter response depends only on ωT modulo 2π.

In western movies, the 24 Hz sampling camera records wagon wheels rotating too slowly or in the retrograde direction.

Digital filters cannot distinguish between sampled signals of frequencies congruent modulo the sampling frequency.

sample at a higher frequency

Exercise 3:

Make the Bode plots of the averaging filter for $T = 1/2000$.

Compare the Bode plots for $T = 1/2000$ and $1/1000$.

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decibels

Relative power levels are expressed in *decibels* (dB):

$$\frac{p_2}{p_1}\ \mathrm{dB} = 10 \log_{10}\left(\frac{p_2}{p_1}\right) = 10 \log_{10}(p_2) - 10 \log_{10}(p_1).$$

Why decibels?

1. To comprehend ratios of vastly different magnitudes.
2. The calculations of gains or losses of cascaded systems are transformed to simple additions and subtractions.

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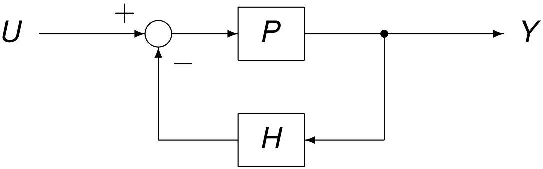
controlling a process

Consider the schematic of a process P:

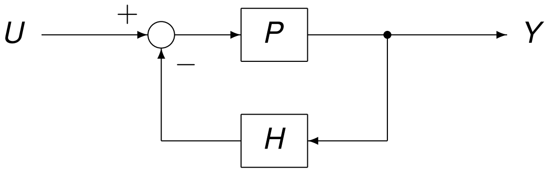
The transfer function of P is $P(z)$, and for z-transforms U (of the input) and Y (of the output), we have: $Y(z) = P(z)U(z)$.

The transfer function of the controller or compensator is H:



The input to P is $U - HY$.

the transfer function of the compensator



The input to P is $U - HY$.

$$\begin{aligned} Y &= P(U - HY) \\ Y &= PU - PHY \\ (1 + PH)Y &= PU \end{aligned}$$

Then we obtain

$$Y = \left(\frac{P}{1 + PH} \right) U.$$

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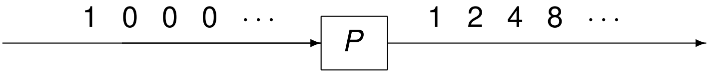
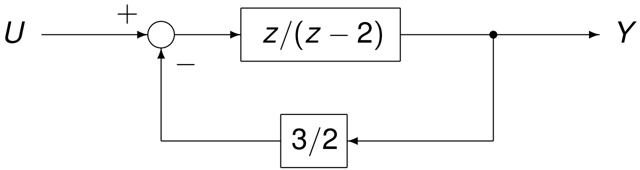
stabilize an unstable plant



The transfer function of P is

$$P(z) = \frac{z}{z-2} = \frac{1}{1-\frac{2}{z}} = 1 + \frac{2}{z} + \frac{4}{z^2} + \frac{8}{z^3} + \cdots + \frac{2^k}{z^k} + \cdots$$

The plant is stabilized by $H(z) = 3/2$ in the closed loop below:



constant feedback

Plant P and compensator H have transfer functions

$$P(z) = \frac{z}{z-2} = \frac{1}{1 - \frac{2}{z}} \quad \text{and} \quad H(z) = 3/2.$$

The transfer function of the closed loop is then

$$\begin{aligned}
\frac{P}{1+PH} &= \frac{1 - \frac{2}{z}}{1 + \frac{3}{2}\left(1 - \frac{2}{z}\right)} = \frac{1}{1 + \frac{2}{z} + \frac{3}{2}} = \frac{1}{\frac{5}{2} - \frac{2}{z}} \\
&= \frac{\frac{2}{5}}{1 - \frac{4}{5}z^{-1}} \\
&= \frac{2}{5}\left(1 + \frac{4}{5}z^{-1} + \left(\frac{4}{5}\right)^2 z^{-2} + \cdots\right)
\end{aligned}$$

As $k \rightarrow \infty$, $\left(\dfrac{4}{5}\right)^k \rightarrow 0$, so the closed loop system is stable.

proportional feedback

Exercise 4:
Show that an unstable plant with transfer function

$$P(z) = \frac{1}{1 - \alpha z^{-1}}, \quad |\alpha| > 1,$$

is stabilized by proportional feedback with transfer function

$$H(z) = \beta$$

for any $\beta > |\alpha| - 1$.